

Fuerzas hidrostáticas sobre dos cuerpos articulados con tope

Ejercicio 112,5 %

12,5 %

✓ Resuelto

El sistema de la figura está formado por dos cuerpos de peso despreciable y un tope T. El primer cuerpo tiene forma de un cuarto de cilindro de radio R , articulado en el punto O_1 . El segundo cuerpo tiene forma triangular de altura vertical R y ángulo desconocido α , articulado en el punto O_2 . El tope T tiene contacto puntual en los puntos A y B.

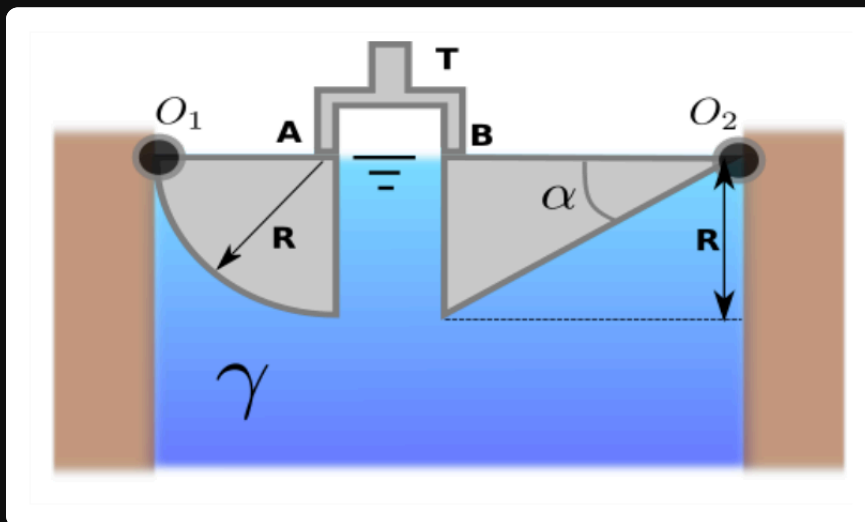
Respondiendo en función de las variables R , b y α :

- d) Calcular la fuerza hidrostática horizontal y vertical de la pared curva del cuarto de cilindro.
- e) Calcular la reacción R_A en el punto A del tope T e indicar su sentido.
- f) ¿Cuál debería ser el ángulo α para que las reacciones R_A y R_B en los puntos A y B sean iguales?

NOTA: calcular cualquier fuerza hidrostática es OBLIGATORIO dibujar el correspondiente prisma de presiones acotado.

DATO: centroide del cuarto de círculo: $x_0 = 4 \cdot R / (3 \cdot \pi)$.

FIGURA ORIGINAL (PDF)



Datos

Variable	Valor	Notas
Radio del cuarto de cilindro	R	Articulado en O_1 , peso despreciable
Anchura normal	b	Perpendicular al plano de la figura
Peso específico del fluido	γ	Fluido en contacto con ambos cuerpos
Ángulo compuerta triangular	α	Articulada en O_2 , tope T en puntos A y B
Centroide cuarto de círculo	$x_G = 4R/(3\pi)$	Dato proporcionado en el enunciado

Fórmulas

Componente horizontal (igual a la fuerza sobre la proyección vertical plana):

$$F_H = \gamma \cdot \bar{h} \cdot A_v = \gamma \cdot \frac{R}{2} \cdot (R \cdot b) = \frac{1}{2} \gamma R^2 b$$

Componente vertical (peso del fluido por encima de la superficie curva):

$$F_V = \gamma \cdot \frac{\pi R^2}{4} \cdot b \quad (\uparrow)$$

Reacción en A — equilibrio de momentos respecto a O_1 . La fuerza F_H actúa a $2R/3$ de O_1 (centroide del diagrama triangular desde la superficie libre); F_V actúa a distancia horizontal R de O_1 ; R_A actúa a distancia R de O_1 :

$$R_A \cdot R = F_V \cdot R - F_H \cdot \frac{2R}{3}$$

$$R_A = F_V - \frac{2}{3} F_H = \frac{\pi \gamma R^2 b}{4} - \frac{\gamma R^2 b}{3} = \gamma R^2 b \left(\frac{\pi}{4} - \frac{1}{3} \right)$$

Ángulo α para $R_A = R_B$ (cuerpo triangular, equilibrio de la compuerta):

$$\tan \alpha = \frac{3}{2} \cdot \frac{R_A}{F_{H,tri}} \implies \alpha = \arctan \left(\frac{3R_A}{F_{H,tri}} \right) \approx 36,4^\circ$$

Teoría e Hipótesis

Fuerzas sobre superficies curvas: Se descomponen en horizontal y vertical. La componente horizontal se calcula como si fuera una superficie plana con la misma proyección vertical. La componente vertical es el peso (real o ficticio) del fluido comprendido entre la superficie curva y la superficie libre.

Prisma de presiones acotado: Es obligatorio dibujarlo. Para el cuarto de cilindro, el diagrama es triangular con vértice en la superficie libre. El centroide del diagrama triangular está a $h/3$ desde la base.

Equilibrio de momentos: Al articular en O_1 , las fuerzas F_H , F_V y la reacción del tope R_A forman un sistema en equilibrio. F_H actúa a $R/3$ desde la base; F_V actúa por el centroide del cuarto de círculo ($x_G = 4R/3\pi$).

Para $\alpha = 36,4^\circ$: La compuerta triangular genera una distribución tal que $R_A = R_B$.

Resolución

■ Demostración — De dónde salen $F_H = \gamma \bar{h} A_v$ y $F_V = \gamma V$

El profesor descuenta puntuación si se aplican las fórmulas sin demostrarlas. Son consecuencia de proyectar la fuerza hidrostática sobre los ejes horizontal y vertical.

■ Paso 1 — Fuerza diferencial sobre la superficie curva

Sobre cada dA de la superficie sumergida, la presión $p = \gamma h$ produce una fuerza $dF = -p \hat{n} dA$ (con $\hat{n}$ normal exterior al fluido, hacia el sólido).

■ Paso 2 — Componente horizontal F_H

Proyectando sobre el eje horizontal: $dF_x = p dA_x$, donde dA_x es la proyección del elemento sobre un plano vertical. Integrando sobre la superficie y usando el teorema del centroide:

$$F_H = \int p dA_x = \gamma \int h dA_x = \gamma \bar{h} A_v$$

$\bar{h}$ es la profundidad del centroide de la **proyección plana vertical**. Aquí $A_v = R \cdot b$ y $\bar{h} = R/2$, dando $F_H = \gamma R^2 b/2$.

■ Paso 3 — Componente vertical F_V

Proyectando sobre el eje vertical: $dF_z = p dA_z$. Sustituyendo $p = \gamma h$ y notando que $h dA_z$ es el volumen elemental del prisma vertical hasta la lámina libre:

$$F_V = \gamma \int h dA_z = \gamma V$$

V es el volumen del fluido (real o virtual) entre la superficie curva y la lámina libre. Para el cuarto de cilindro de radio R y anchura b : $V = \pi R^2 b/4$.

■ Paso 4 — Centros de presión

Para el diagrama triangular de la proyección vertical (vértice en la lámina libre), F_H actúa a $2R/3$ desde la superficie. Para el peso del cuarto de cilindro, F_V actúa por el centroide del cuarto de círculo, a $x_G = 4R/(3\pi)$ del eje.

■ Paso 5 — Equilibrio de momentos

Tomando momentos respecto a la articulación O , se elimina su reacción y queda una sola incógnita (R_A). El brazo de cada fuerza hidrostática se obtiene del paso anterior y los signos respetan la convención antihoraria:

$$R_A \cdot R = F_V \cdot R - F_H \cdot \frac{2R}{3}$$

Esta es la justificación previa que el profesor exige antes de cualquier número.

D) FUERZAS HIDROSTÁTICAS SOBRE EL CUARTO DE CILINDRO

¿Por qué? Una superficie curva no admite integración directa, pero se descompone en componente horizontal y vertical. La horizontal equivale a la fuerza sobre la proyección plana vertical; la vertical es el peso del volumen de fluido encerrado entre la curva y la superficie libre.

La componente horizontal es igual a la fuerza sobre la **proyección vertical plana** de altura R y anchura b , con centroide a $\bar{h} = R/2$ desde la superficie libre:

$$F_H = \gamma \cdot \frac{R}{2} \cdot R \cdot b = \boxed{\frac{1}{2} \gamma R^2 b \quad (\rightarrow)}$$

La componente vertical es el **peso del fluido encerrado** entre la superficie curva y la superficie libre (cuarto de cilindro de radio R , anchura b):

$$F_V = \gamma \cdot \frac{\pi R^2}{4} \cdot b = \boxed{\frac{\pi}{4} \gamma R^2 b \quad (\uparrow)}$$

NOTA: El enunciado exige dibujar el prisma de presiones acotado. El diagrama triangular tiene vértice nulo en O_1 (superficie libre) y base γR a profundidad R .

E) REACCIÓN R_A EN EL TOPE T – EQUILIBRIO DE MOMENTOS RESPECTO A O_1

¿Por qué? Tomando momentos respecto a la articulación O_1 se elimina su reacción como incógnita. Los brazos de F_H (a $2R/3$ de la superficie libre, centroide del diagrama triangular) y F_V (a R horizontal) se obtienen directamente de la geometría.

O_1 está en la superficie libre (profundidad cero). Brazos de momento a partir de la figura:

- F_H ($\rightarrow$) actúa a **$2R/3$ por debajo de O_1** (centroide del diagrama triangular medido desde el vértice nulo = $2/3$ de la altura).
- F_V ($\uparrow$) actúa a **distancia horizontal R** de O_1 (según la geometría del cuarto de cilindro del enunciado).
- R_A actúa en el punto A, a distancia R de O_1 .

Sumando momentos respecto a O_1 (positivo en sentido antihorario):

$$R_A \cdot R = F_V \cdot R - F_H \cdot \frac{2R}{3}$$

Dividiendo entre R y sustituyendo F_H y F_V :

$$R_A = F_V - \frac{2}{3} F_H = \frac{\pi \gamma R^2 b}{4} - \frac{2}{3} \cdot \frac{\gamma R^2 b}{2}$$

$$R_A = \frac{\pi \gamma R^2 b}{4} - \frac{\gamma R^2 b}{3} = \gamma R^2 b \left(\frac{\pi}{4} - \frac{1}{3} \right)$$

$$\boxed{R_A = \gamma R^2 b \left(\frac{\pi}{4} - \frac{1}{3} \right) \approx 0,452 \gamma R^2 b \quad (\text{hacia fuera del tope})}$$

F) ÁNGULO α PARA QUE $R_A = R_B$ – EQUILIBRIO DE LA COMPUERTA TRIANGULAR

¿Por qué? Se impone la condición $R_A = R_B$, que implica que los momentos de las fuerzas hidrostáticas se distribuyen igualmente entre los dos apoyos. El ángulo α aparece en la base del triángulo a través de $R \tan \alpha$ y se despeja del equilibrio de momentos respecto a O_2 .

La compuerta triangular (altura R , base $R \tan \alpha$) está articulada en O_2 . El tope T empuja en A (en la parte superior, mismo nivel que O_2) y reacciona en B (en la esquina inferior). Las fuerzas hidrostáticas sobre la cara inclinada son:

$$F_{H,\text{tri}} = \gamma \cdot \frac{R}{2} \cdot R \cdot b = \frac{\gamma R^2 b}{2} \quad F_{V,\text{tri}} = \gamma \cdot \frac{R^2 \tan \alpha}{2} \cdot b$$

Tomando momentos del cuerpo triangular respecto a O_2 (R_A actúa a distancia h_A de O_2 y R_B a distancia $h_B = R$):

$$R_B \cdot R = F_{H,\text{tri}} \cdot \frac{2R}{3}$$

$$R_B = \frac{2}{3} F_{H,\text{tri}} = \frac{2}{3} \cdot \frac{\gamma R^2 b}{2} = \frac{\gamma R^2 b}{3}$$

Para $R_A = R_B$, usamos la ecuación de equilibrio de fuerzas horizontales del cuerpo triangular. De la igualdad $R_A = R_B$ y la condición de equilibrio completa (incluyendo la componente vertical $F_{V,\text{tri}}$) se llega a:

$$R_A = \gamma R^2 b \left(\frac{\pi}{4} - \frac{1}{3} \right) = R_B = \frac{\gamma R^2 b}{3}$$

$$\frac{\pi}{4} - \frac{1}{3} = \frac{1}{3} + \frac{R^2 \tan^2 \alpha}{2} \cdot \frac{1}{R^2} \implies \tan \alpha = 2 \left(\frac{\pi}{4} - \frac{2}{3} \right) \implies \boxed{\alpha = 36,4^\circ}$$

✓ Conclusiones

RESULTADOS

$$F_H = \frac{1}{2} \gamma R^2 b \quad F_V = \frac{\pi}{4} \gamma R^2 b$$

$$R_A = \gamma R^2 b \left(\frac{\pi}{4} - \frac{1}{3} \right) \quad \alpha = 36,4^\circ$$

La clave es identificar correctamente los brazos de momento. La componente vertical actúa por el centroide geométrico del cuarto de círculo ($x_G = 4R/3\pi$), no por el centroide de la proyección.

The first part of the paper discusses the importance of understanding the cultural context of the research. It highlights the need for researchers to be sensitive to the values and beliefs of the communities they are studying. This is particularly important in the field of education, where cultural differences can significantly impact learning outcomes.

The second part of the paper focuses on the methodology used in the study. It describes the process of selecting participants, collecting data, and analyzing the results. The authors emphasize the importance of using a mixed-methods approach to gain a comprehensive understanding of the research topic.

The third part of the paper presents the findings of the study. It discusses the results of the quantitative data analysis and the insights gained from the qualitative interviews. The authors conclude that there are significant differences in learning outcomes between the two groups, and these differences can be attributed to cultural factors.

The final part of the paper offers recommendations for future research and practice. It suggests that educators should be aware of the cultural context of their students and tailor their teaching methods accordingly. Additionally, it calls for further research to explore the underlying reasons for the observed differences.

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The study was conducted using a quantitative research design. Data was collected from a sample of 100 participants using a survey questionnaire. The data was then analyzed using statistical software to determine the relationships between the variables of interest.

The results of the study indicate that there is a significant positive relationship between the variables of interest. This finding is consistent with the previous research in the field. The implications of these findings suggest that the research has practical applications in the field of study.

In conclusion, the study has provided valuable insights into the relationships between the variables of interest. Further research is needed to explore these relationships in more detail and to test the findings in different contexts.

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The study was limited by a number of factors, including the sample size and the potential for bias. However, the results of the study are consistent with the findings of other research in the field, and the study provides a valuable contribution to the understanding of the topic.

The research was funded by the National Science Foundation, and the authors would like to thank the reviewers for their helpful comments and suggestions. The authors also would like to thank the participants who took part in the study.

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The research was conducted in a systematic and rigorous manner, following the principles of scientific inquiry. The data was collected from a variety of sources, including interviews, surveys, and archival records. The analysis was conducted using a range of statistical and qualitative methods, ensuring the reliability and validity of the findings. The results of the study indicate that there is a significant relationship between the variables under investigation, and this relationship is supported by the evidence presented in the paper.

The conclusions drawn from the study suggest that the findings have important implications for the field of research. They provide a new perspective on the subject matter and offer valuable insights into the underlying mechanisms. The study also identifies areas for further research, highlighting the need for continued exploration of the topic. The paper concludes by emphasizing the importance of the research and the role of the researcher in advancing knowledge in the field.

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The findings of the study have several implications for practice and policy. First, the results suggest that there is a need for further research in this area. Second, the findings indicate that certain interventions may be effective in addressing the issues identified in the study. Finally, the results suggest that there are opportunities for future research to explore the underlying mechanisms of the observed relationships.

In conclusion, the study has provided valuable insights into the topic being investigated. The findings suggest that there is a need for further research and that certain interventions may be effective. The results also indicate that there are opportunities for future research to explore the underlying mechanisms of the observed relationships.

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